Topic :Semantic Error Detection

By: Ojas Jogdand
(23BDS039)

Dharwad

ज्ञानेन विकासः

SURVEY OF KEY RESEARCH PAPERS

Research Papers — Method Comparison

Semantic Error Detection

Paper	Core Method	Dataset	Task	Metric	Fine-tune?
Warstadt et al. (2019)	BERT / GPT fine-tune	CoLA (10,657)	Acceptability classif.	MCC	Yes ✓
Nihalani & Shah (2024)	BERT / RoBERTa / GPT-4	Lang-8 cleaned	Binary GED	F1, Acc.	Yes / No
Zhang et al. (2025)	Bidirectional GEC+CoLA	CoLA + GEC corpora	Acceptability+GEC	MCC+BLEU	Yes ✓
Yuan et al. (2022)	Semantic clustering (iSEA)	Various NLP tasks	Error diagnosis	Error rate	No
Lu et al. (2024)	EAPrompt (CoT)	WMT translations	Translation eval.	MQM / human	No
This Project ★	**BERT-base fine-tuned & tested**	**Trans. Error + CoLA**	**Semantic error det.**	**MCC**	**Yes ✓**

SURVEY OF EXISTING APPROACHES

1. Rule-Based | Pre-2010
   - High precision on known patterns
   - Zero recall on unseen errors
   - No contextual understanding

2. Statistical / n-gram | 2010–2015
   - Uses perplexity & co-occurrence (PMI)
   - Broader coverage than rules
   - Cannot capture global meaning

SURVEY OF EXISTING APPROACHES

3. Static Embeddings (Word2Vec / GloVe) | 2014–2018
    - Efficient and captures semantic similarity
    - One vector per word → no context awareness
    - Fails on polysemy

4. Fine-tuned Transformers | 2019–Present
    - Strong performance (e.g., MCC ≈ 0.605, Accuracy ≈ 0.834)
    - Task-specific learning
    - Requires labeled data + GPU

DATA PIPELINE & PREPROCESSING

Training Dataset:
English Translation Error Detection Dataset
https://www.kaggle.com/datasets/zara2099/english-translation-error-detection-dataset

Testing Dataset:
CoLA(Corpus of Linguistic Acceptability)
https://www.kaggle.com/datasets/krazy47/cola-the-corpus-of-linguistic-acceptability

Preprocessing Steps:

- Selected relevant columns: sentence + label
- Converted labels to binary format (0/1)
- Removed null values

Hard Negative Generation:

- Created semantically incorrect versions of each sentence
- Example:
- He deposited money at the bank
- He deposited money at the park

Data Split: 80% training,10% validation,10% testing

MODEL ARCHITECTURE

Model Used :BERT + Contrastive Learning Model

Base Model:

- BERT (bert-base-uncased)
- Extracts contextual sentence representation

Model Components:

- CLS token → sentence embedding
- Fully connected layer → classification (Correct / Error)

Extended Output:

- Logits → classification output
- CLS embedding → semantic representation

Training Strategy(Confidence-Calibrated Contrastive Learning):

- Classification Loss (Improved):
  Confidence-weighted cross entropy and Focuses more on difficult examples
- Contrastive Loss:
  Forces: Correct sentence embedding ≠ incorrect sentence embedding

ANALYSIS

Model	Accuracy	Precision	Recall	F1-score
BERT(Baseline)	0.7259	0.7520	0.7335	0.6120
BERT+CCCL	0.7430	0.7605	0.7430	0.5917

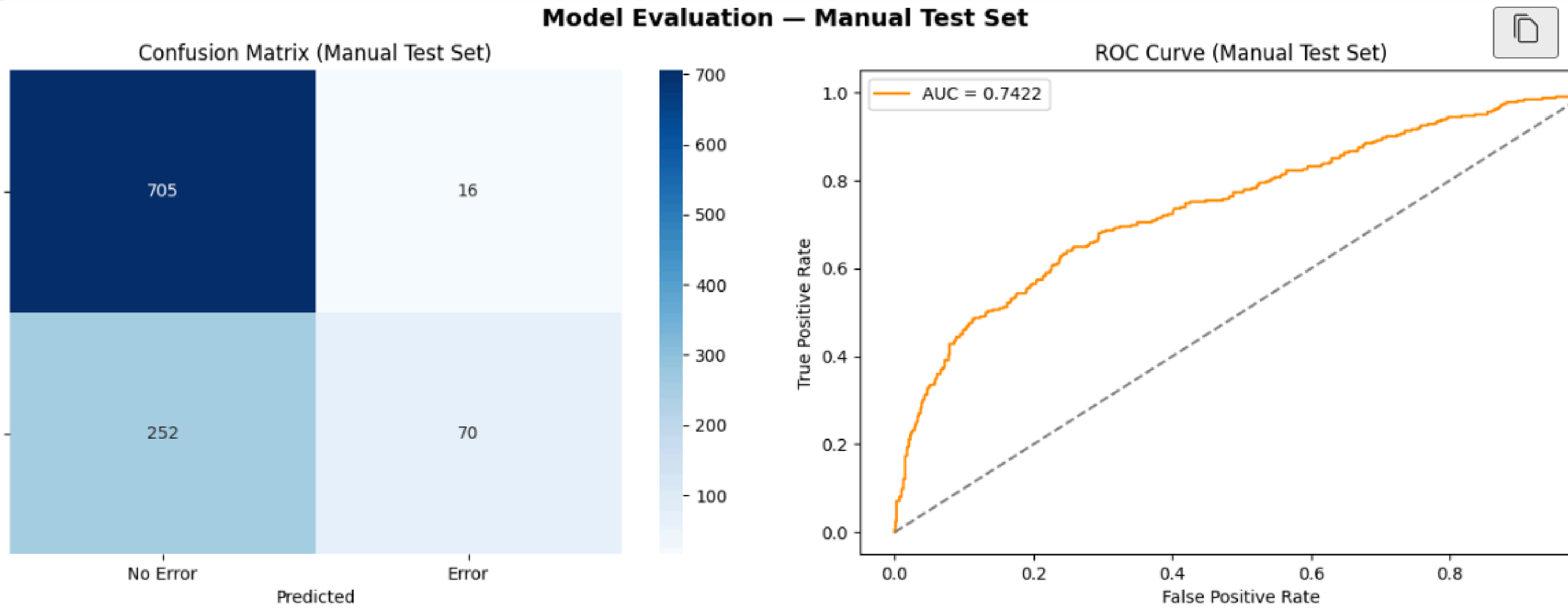
- Overall performance improvement is modest (+1–2%)
- However, CCCL improves semantic sensitivity and handling of confusing sentence pairs

Example:

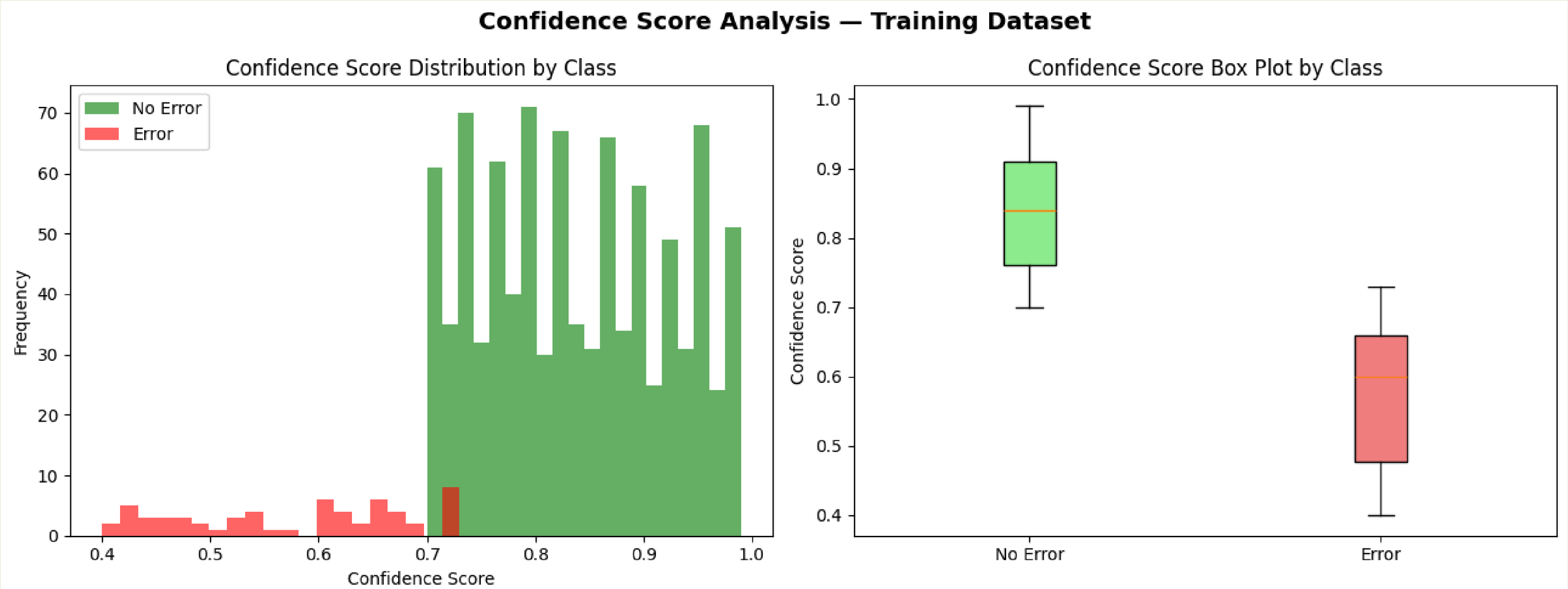
Without CCCL:
"He deposited money at the park" → predicted correct

With CCCL:
correctly identified as error

ANALYSIS


Model Evaluation — Manual Test Set
Confusion Matrix (Manual Test Set)
705
16
252
70
No Error
Error
Predicted
700
600
500
400
300
200
100
ROC Curve (Manual Test Set)
AUC = 0.7422
True Positive Rate
False Positive Rate
0.0
0.2
0.4
0.6
0.8
1.0

ANALYSIS


Confidence Score Analysis — Training Dataset
Confidence Score Distribution by Class
No Error
Error
Frequency
Confidence Score
0
10
20
30
40
50
60
70
0.4
0.5
0.6
0.7
0.8
0.9
1.0
Confidence Score Box Plot by Class
Confidence Score
0.4
0.5
0.6
0.7
0.8
0.9
1.0
No Error
Error

INNOVATION

Problem in Existing Models:
Standard BERT:

- Classifies sentences as correct / incorrect
- Struggles with subtle semantic errors
- Fails when sentences are very similar in wording

My Solution: CCCL (Confidence-Calibrated Contrastive Learning)
BERT is enhanced by introducing:

- Hard negative samples
- Contrastive learning
- Confidence-weighted loss

Key Objective:

- Improve semantic understanding
- Detect small meaning differences
- Focus on difficult examples

INNOVATION

1.Hard Negative Generation

- Create slightly incorrect versions of sentences
- Example
  Instead of “bank” → “park”
- Impact: Helps model detect subtle semantic error

2.Contrastive Learning

- Model Compares:
  1)Original Sentence 2)Hard negative sentence
- Impact:
  1) Keeps the embeddings of 2 similar sentences far away.
  2) Learns difference between similar sentences
  3) Improves semantic representation

3.Confidence-Weighted Loss

- Gives more importance to Low-confidence (hard) examples
- Impact:
  1) Better learning on confusing cases
  2)Reduces overfitting to easy examples

INNOVATION

Why This Innovation Matters

Without CCCL:

- Focuses only on classification
- Misses subtle meaning errors
- Treats all samples equally

With CCCL:

- Learns semantic differences explicitly
- Separates similar sentences in embedding space
- Focuses on hard-to-classify examples

Overall Benefit:

- Improved semantic sensitivity
- Better detection of real-world translation errors

LIMITATIONS

- Small improvement in metrics
  ->Task is inherently difficult (semantic understanding)
- Simple hard negative generation
  ->May not capture real-world errors perfectly
- Higher computation cost
  ->Two inputs per training step
- Still struggles with:
  ->Complex sentence structures
  ->Rare or unseen vocabulary

Future work:

- Better negative generation (context-aware)
- Try stronger models (RoBERTa, LLMs)
- Improve semantic reasoning further

CONCLUSION

- Implemented BERT-based semantic error detection
- Introduced CCCL (Contrastive + Confidence Learning)
- Added:
  ->Hard negative samples
  ->Contrastive learning
  ->Confidence-weighted loss

Results:

- Slight improvement (~1–2%) in Accuracy & F1
- More importantly:
  ->Better detection of subtle semantic errors
  ->Improved handling of similar-looking sentences